What Reaches the Model? Trace-Based Diagnosis and Repair of Component Faults in RAG Pipelines

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Abstract

Retrieval-augmented generation (RAG) services fail not only in retrieval and generation but in the code between them: templates truncate passages, serving runtimes silently cut prompts that exceed their context window, indexes keep outdated document versions, and query encoders drift from the encoder the index was built with. We inject six such faults into a complete RAG service running on a laptop and compare twelve runtime repair policies, under a budget of two extra model calls, on 150 HotpotQA questions with Llama-3.2-3B. Generic repairs fail: retry barely changes success under faults (26.8% to 27.6%), and a self-check with the model's own verifier lowers it to 20.7% while breaking 24.3% of correct answers. Applying every repair on every request reaches 46.4% but breaks 15.9% of correct answers and makes a serving fault worse than doing nothing. Trace checks, computed from data the service already logs (rendered versus retrieved text, tokens sent versus tokens processed, duplicate document versions, retrieval confidence), recover 55.9% with 1.9% damage, and a diagnose policy built on them reaches 56.9% (+30.1 points over no repair, 95% CI [24.8, 35.7]). Whether an answer is judged unsupported by the model itself, by a MiniCheck grounding classifier, or by text match matters far less than whether the repair fixes the context. In simulated request streams, trace checks detect a faulty deploy a median of 4 requests after it ships; model-based signals need tens of requests or never alarm. A regression suite built from observed failures replays in seconds from recorded responses and fails whenever a diagnostic is disabled.

Keywords

retrieval-augmented generation, fault injection, runtime monitoring, repair, ML in production

1 Introduction

A retrieval-augmented generation (RAG) service is a pipeline: a retriever finds passages, glue code assembles them into a prompt, a serving runtime runs the model, and an index holds the documents. Evaluation frameworks for RAG score the retrieved context and the final answer [6, 21, 22]. In production, however, evidence can be lost *between* those two points. A government help-desk assistant lost its system prompt and its top-ranked documents for weeks because the serving runtime silently kept only the tail of a prompt that exceeded its default context window [16]; users of a popular RAG platform report the same problem when the model server ignores the configured window [19]. Template code that truncates passages, an index that still holds an outdated version of an updated document [17], and a query encoder that no longer matches the encoder the index was built with are similar failures: the retriever may work, the model may work, and the answer is still wrong.

When answers look wrong, a common runtime response is generic: retry, or ask the model whether its answer is supported and regenerate if not. Such repairs spend calls on the generator even when the generator is not the component that failed, and LLMs are known to be unreliable at verifying their own outputs without external feedback [9]. This paper asks a practical question for teams operating RAG services: *when a pipeline component fails, which runtime repair helps, what does it break, and what does it cost?*

We study this with a fault-injection harness around a small but complete RAG service running locally. We inject six faults motivated by reported incidents at four pipeline stages (retrieval, prompt assembly, serving, index content), and compare twelve repair policies under a fixed budget of two extra model calls. The policies range from generic (retry, self-check) to *trace-based*: checks computed from what the service already logs, such as whether the rendered prompt is shorter than the retrieved text or whether the server processed fewer tokens than were sent. We measure success under faults, damage to answers that were correct, abstention when the evidence is gone, and cost. We address four questions:

- **RQ1** How do generic, blanket and trace-based repairs compare under component faults?
- **RQ2** Does the verifier or the repair action matter more? We cross {LLM self-verifier, MiniCheck grounding classifier [24], text match} with {regenerate on the same passages, repair the context}.
- **RQ3** How quickly do trace checks and model-based signals detect a faulty deploy in a request stream?
- **RQ4** Can a regression suite built from observed failures catch regressions in the repair logic itself?

On 150 held-out HotpotQA questions with Llama-3.2-3B, trace-based diagnosis raises success under faults from 26.8% to 56.9% (+30.1 points, 95% CI [24.8, 35.7]) while breaking 4.7% of correct clean answers; trace checks alone account for almost all of the gain. A generic LLM self-check lowers success to 20.7% and breaks 24.3% of correct clean answers. Applying every repair on every request reaches 46.4%, breaks 15.9% of correct clean answers, and makes the serving fault worse than doing nothing (6.0% vs. 30.0%). Trace checks detect each global fault a median of 4 requests after a deploy; the model's own verifier needs tens of requests and often never alarms. A 35-case regression suite, replayed from recorded responses in seconds, fails whenever one of the diagnostics is disabled.

Contributions. (1) A fault-injection study of RAG runtime repair that treats prompt assembly and serving as components that can fail; (2) a budget-matched comparison of twelve repair policies that accounts for damage and abstention; (3) evidence that cheap,

model-free trace checks recover most of the benefit and detect faulty deploys fastest, while model self-verification is harmful at this scale; and (4) an open, laptop-scale artifact with record/replay regression tests for LLM pipelines.

2 Related Work

RAG failures and evaluation. Barnett et al. catalog seven failure points of RAG systems and argue that RAG can only be fully validated in operation [2]. RAGAS, ARES and RAGChecker score retrieval and generation quality [6, 21, 22]; RAGChecker separates retriever and generator errors with claim-level metrics. Chen et al. study 654 configuration issues in ten RAG systems and find that root causes often sit in a different stage than the symptoms [4]. Kim et al. apply metamorphic testing to RAG under corpus changes [12]. These works measure the retrieved context and the answer; we additionally treat prompt assembly and serving, the glue code that Sculley et al. identify as a source of hidden debt [23], as components that can fail, and we test at runtime rather than offline.

Repair and self-correction. Self-RAG trains a model to critique its retrievals and generations [1]; CRAG uses a retrieval evaluator to trigger corrective retrieval [26]; Doctor-RAG localizes failures in agentic RAG trajectories and repairs from the earliest failure point [10]. Huang et al. show that LLMs cannot reliably self-correct reasoning without external feedback [9]. We compare lightweight, training-free policies under a fixed call budget and measure what they break.

Grounding, staleness and abstention. MiniCheck provides small, accurate grounding classifiers [24]. HoH shows that outdated information coexisting with current information degrades RAG [17]. RefusalBench evaluates when grounded models should refuse [15].

ML in production. The ML Test Score lists monitoring and testing practices for production ML [3]; Ma et al. argue for slice-based regression tests of evolving LLM APIs [14]; Dutta et al. show that aggregate accuracy hides changed predictions after model compression [5], the motivation for our clean-damage measure. Our monitoring and record/replay tests follow the practices described by Kästner [11].

3 Study Design

3.1 System under test

The service answers a question in three steps. (1) *Retrieval*: dense retrieval with `bge-small-en-v1.5` [25] over a fixed passage store, top k=5. A BM25 index over the same store is available as a lexical fallback. (2) *Prompt assembly*: each passage is rendered as `[Pi] title (last updated date)` followed by its text, and the instruction asks for the shortest possible answer plus the ids of supporting passages, or `INSUFFICIENT`. (3) *Serving*: a 4-bit instruction-tuned model served locally with MLX [8], greedy decoding. The service logs a trace for every call: retrieved documents with their similarity scores and metadata, the passages exactly as rendered, the number of prompt tokens it sent and the number the server reports as processed, and the raw output. Repair policies see this trace and nothing else; they never see the fault label or the gold answer.

Table 1: Injected faults, by pipeline stage.

Fault	Injection	Production cause
encoder	queries embedded with `all-MiniLM-L6-v2` [20]; index built with `bge-small`	query service falls back to a library default encoder
truncation	template cuts every passage to its first 100 characters	`text[:max_chars]` misconfiguration
overflow	server keeps only the last 512 prompt tokens	unset context window, e.g. `num_ctx` default [16, 19]
stale	old version of the answer passage indexed beside the current one; retrieval ranks both	index not deduplicated after an update [17]
absent	answer passage removed from the index	content gap
enc.+trunc.	encoder and truncation together	independent faults co-occur

3.2 Data

We use the HotpotQA distractor validation set [27]. We keep *bridge* questions whose answer string occurs in exactly one of the question's ten paragraphs, in a gold supporting sentence, and drop answers with annotation artifacts. The first 1,200 such questions (seeded shuffle) define the store: all their paragraphs, deduplicated by title (11,737 passages). Of these, 664 have both gold paragraphs in the clean dense top-5; this filter guarantees that in the clean condition the evidence is present, so each injected fault is the only reason it goes missing. We draw 30 development and 150 test questions and exclude every question used in an earlier prototype of this study. Each passage carries a "last updated" date (2023–2024). For the stale fault, each question's answer passage also has an *old version*, dated three to six years earlier, in which the answer value is replaced by a plausible wrong value: numbers and dates are shifted by rule; entity answers are rewritten by a local 7B model (e.g., Sam Waterston → Tom Hanks) and checked to differ from the gold answer.

3.3 Faults

Table 1 lists the six faults. Each models a failure reported for deployed RAG services, and none targets the answer passage directly except *absent*, where abstaining is the correct behavior.

3.4 Signals and repair policies

Trace checks are computed from the trace with no model call: (i) *truncation*, a rendered passage is shorter than the retrieved text; (ii) *version conflict*, two passages share a title; (iii) *overflow*, the server processed fewer prompt tokens than were sent; (iv) *low retrieval*, the top dense similarity is below the 5th percentile of clean traffic, calibrated on 467 held-out questions. Each check has a matching repair: render in full, keep the newest version, fit the passages to the window the server reported, or fall back to BM25. *Support checks* decide whether an answer is grounded: a text-match check (the answer appears in the passages), the model's own verifier ("Is the proposed answer correct and directly supported by the passages? SUPPORTED/UNSUPPORTED"), or MiniCheck-RoBERTa-Large [24], a

Table 2: Repair policies (budget: two extra model calls).

Policy	Behavior
No repair	deployed pipeline
Retry	if `INSUFFICIENT`, resample at temperature 0.7
Self-check (2 verifiers)	verify the answer (LLM or MiniCheck); if unsupported, regenerate on the same passages with a hint
Repair-all	always: dense top-10 ∪ BM25 top-5, full rendering, newest versions, regenerate
Trace checks only	trace checks and their repairs; no support check
Diagnose (3 support checks)	trace checks, then a support check (text match, LLM, or MiniCheck) with re-check after each repair, retrieval ladder, abstention
ablations	proposed (text match) variant without re-check, without abstention, without trace checks

355M-parameter grounding classifier run on the passages packed into 400-token chunks. MiniCheck's model and claim format were chosen on prototype development data; its thresholds were set on the development split by rules fixed in advance (0.5, MiniCheck's default, to trigger a repair; the 2nd percentile of scores of correct grounded answers, 0.01, to abstain). The support check of the proposed policy (text match) was also chosen on the development split; the other two remain as comparison arms.

Table 2 lists the policies. All start from the same initial answer and may make at most two more model calls; LLM verifier calls count against this budget, MiniCheck calls do not (their time is reported separately). *Diagnose* applies the repairs indicated by the trace checks, then checks support; while the answer is unsupported and budget remains it widens retrieval to 10, then adds BM25 results, regenerating after each step; if the final answer is still clearly unsupported it returns `INSUFFICIENT`. The self-check and diagnose variants form a factorial design: {LLM verifier, MiniCheck} × {regenerate on the same passages, repair the context}, plus text match with context repair.

3.5 Measures and statistics

Success is a correct answer in the answerable conditions and `INSUFFICIENT` in *absent*. Correctness is exact match after normalization (number words mapped to digits), token F1 ≥ 0.8, or the gold phrase inside a short prediction; other predictions go to a local answer-equivalence judge (Qwen2.5-7B) that sees only the question, gold answer and prediction. On the prototype, 197 of 203 audited judge decisions agreed with a manual audit; we report results with string matching alone as a sensitivity check. *Success under faults* averages the five answerable fault conditions per question. *Clean damage* is the share of clean questions answered correctly at first and wrongly after the policy [5]. Costs are model calls, prompt tokens and GPU seconds per case. All comparisons are paired by question: 95% intervals are percentile bootstraps over questions (2,000 resamples), per-condition tests are exact McNemar.

3.6 Model and setup

The generator is Llama-3.2-3B-Instruct [13], 4-bit, served with MLX on one MacBook Pro (M4, 16 GB); this paper reports one model, and extending the study to Qwen2.5-7B [18] and Gemma-2-9B [7] is ongoing (Section 6). A response cache keyed by model, prompt and sampling settings lets policies that issue the same prompt share generations and makes every run replayable.

Table 3: Test results (Llama-3.2-3B, 150 questions). Faults: success under the five answerable faults (%). Dmg.: correct clean answers broken (%). Abst.: abstention when the answer passage is absent (%). Tok.: prompt tokens per case.

	Llama-3.2-3B			
Policy	Faults	Dmg.	Abst.	Tok.
No repair	26.8	0.0	38.7	0.7k
Retry	27.6	0.0	34.0	0.9k
Self-check (LLM)	20.7	24.3	60.7	1.5k
Self-check (MiniCheck)	22.4	12.1	55.3	1.0k
Repair-all	46.4	15.9	20.0	2.4k
Trace checks only	55.9	1.9	38.0	1.2k
Diagnose (text match)	56.9	4.7	28.0	1.5k
Diagnose (LLM verifier)	57.2	11.2	28.0	2.3k
Diagnose (MiniCheck)	59.3	7.5	29.3	2.1k
− re-check	53.1	5.6	30.0	1.3k
− abstention	59.9	1.9	23.3	1.5k
− trace checks	31.9	3.7	30.0	1.3k

4 Results

4.1 RQ1: Which repairs help?

Table 3 and Figure 1 summarize the comparison; all differences below are paired by question.

Generic repairs do not help. With no repair the service answers 71.3% of clean questions but only 26.8% [22.7, 31.2] under faults. Retry changes little (27.6%): it only fires on `INSUFFICIENT` and resamples the same broken prompt. Self-check is worse than no repair (20.7%) and breaks 24.3% of correct clean answers, almost always by turning them into `INSUFFICIENT` after its verifier calls them unsupported.

Blanket repair helps but breaks things. Repair-all (wider dense retrieval plus BM25, full rendering, newest versions, on every request) reaches 46.4% under faults. It costs a second call with 3.6× the prompt tokens of the deployed pipeline, breaks 15.9% of correct clean answers, and lowers abstention on unanswerable questions from 38.7% to 20.0%. Under the serving fault it is far worse than doing nothing (6.0% vs. 30.0%): its longer prompt loses even more to the server's silent cut, so the repair amplifies the fault.

Trace checks do most of the work. The four trace checks with their matching repairs reach 55.9% [49.7, 62.0] under faults while breaking 1.9% of correct clean answers, at 1.72 calls and 1.2k prompt tokens per case. Diagnose adds a support check, a retrieval ladder and abstention: 56.9% [50.7, 63.1], +30.1 points over no repair [24.8, 35.7] and +10.5 over Repair-all [6.4, 14.9], but only +1.1 [−2.0, 4.0] over trace checks alone. The ablations explain why. Removing the trace checks drops success to 31.9% (−25.1 [−30.0, −20.3]): a support check that sees only the answer cannot tell a template bug from a retrieval miss, and widening retrieval does not undo truncation or

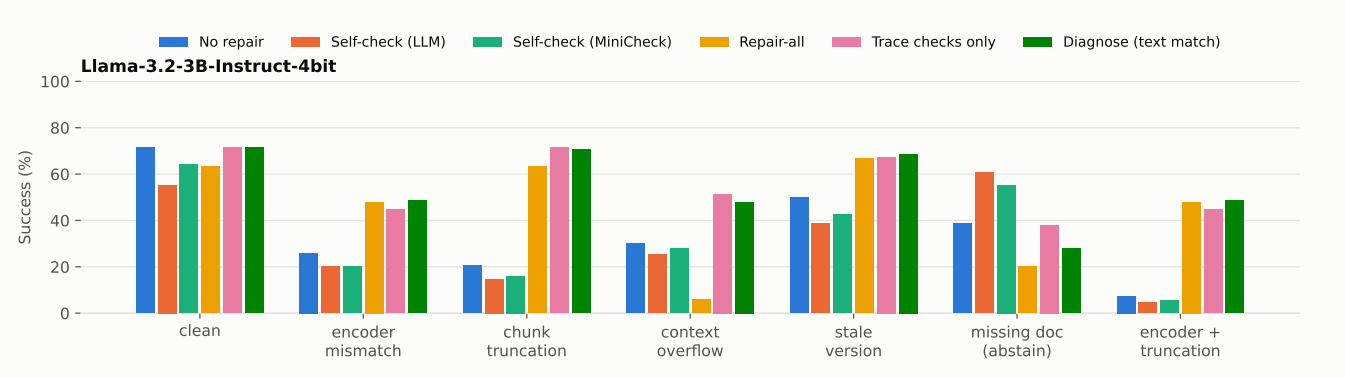


Figure 1: Success by condition for six policies (Llama-3.2-3B, 150 test questions). In *absent*, success means abstaining.

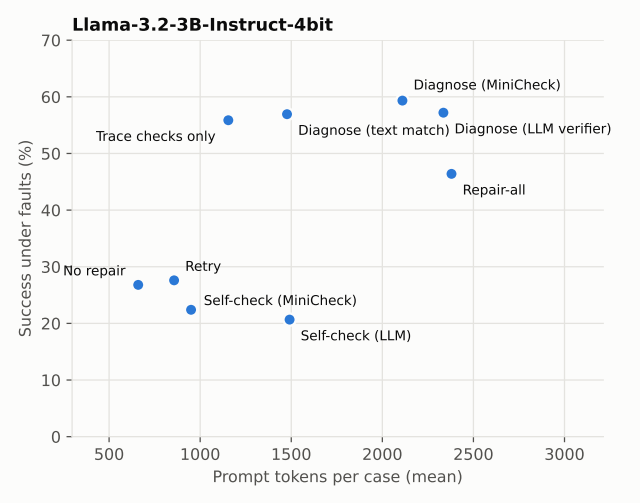


Figure 2: Success under faults versus prompt tokens per case.

overflow. Re-checking support after a repair is worth +3.9 points [1.9, 6.0], mostly under the encoder fault, where the first fallback is often not enough. Abstention costs 2.9 points under faults [1.1, 5.3] and raises clean damage from 1.9% to 4.7%, while recovering only 4.7 points of appropriate abstention.

The checks are precise. The truncation and overflow checks fired on every faulty case and on no clean case; the version check fired on 86% of stale cases (the outdated copy was in the prompt in 91%); the low-retrieval check fired on every encoder-fault case and, by construction of its threshold, on 4% of clean cases, where its BM25 fallback did no measurable harm.

Repairs reduce abstention. When the answer passage is missing, the deployed pipeline abstains 38.7% of the time; Diagnose abstains 28.0% (p=0.004). The retrieval ladder adds passages, and the model then finds a plausible but wrong entity to name. Only the self-check policies abstain more (55–61%), and they pay for it with clean damage.

Table 4: Verifier × repair. Cells: success under faults / clean damage (%).

Model	Verifier	Regenerate	Repair context
Llama-3.2-3B	LLM self-verifier	20.7 / 24.3	57.2 / 11.2
	MiniCheck	22.4 / 12.1	59.3 / 7.5
	Text match	–	56.9 / 4.7

4.2 RQ2: Verifier or repair?

Table 4 crosses the decision of *whether* an answer is unsupported with *what happens next*. Regenerating on the same passages fails with either verifier (20.7%, 22.4%); repairing the context succeeds with any of the three (56.9–59.3%). The repair action, not the verifier, determines recovery. The verifier mainly determines damage: the LLM self-verifier breaks the most correct answers in either column.

Table 5 evaluates the gates directly on 700 non-abstaining initial answers. The LLM self-verifier keeps only 26.3% of correct, grounded answers. MiniCheck is better balanced (AUROC 0.67) but still rejects nearly half of the good answers at its default threshold; multi-hop answers whose support spans two passages are hard for a 355M-parameter grounding classifier. Is this a small-model artifact? On a stratified sample of 240 of the same answers we asked Claude Haiku 4.5 the identical verifier question. It keeps 71.7% of good answers and catches 82.4% of bad ones (balanced accuracy 0.77), against 20.7% and 91.2% (0.56) for the 3B model's self-verifier and 0.63 for MiniCheck. A stronger verifier would remove most of the false alarms that made self-check destroy correct answers. It would not make self-check repair anything: under truncation, overflow or a retrieval fault the evidence is not in the passages, so regenerating on them cannot recover it, which is what the left column of Table 4 shows for both verifiers. Even the strong verifier is close to chance on truncated passages (0.53), where the question is whether missing text would have supported the answer.

4.3 RQ3: How fast is a faulty deploy detected?

We simulate 300 request streams per fault: 400 clean requests, then 200 after a deploy that introduces the fault (the stale fault affects 20% of post-deploy requests, the others all of them). A signal alarms when its rate over the last 50 requests exceeds the clean mean

Table 5: Support checks as gates on initial answers. Good kept: correct, grounded answers judged supported. Bad caught: other answers judged unsupported.

Check	Good kept	Bad caught	Bal. acc.
Llama-3.2-3B: 700 initial answers, 285 correct and grounded			
LLM self-verifier	26.3	86.7	0.57
MiniCheck ≥ 0.5 (AUROC 0.67)	52.6	72.8	0.63
Same 240 answers (40 per condition) for all three			
Llama-3.2-3B self-verifier	20.7	91.2	0.56
Claude Haiku 4.5, same prompt	71.7	82.4	0.77
MiniCheck ≥ 0.5	45.7	81.1	0.63

Table 6: Detection delay after a faulty deploy: median requests until the first alarm (share of 300 runs with an alarm). –: detected in fewer than half the runs.

Fault	Trace	MiniCheck	LLM ver.	Abstain
encoder	4 (100%)	23 (100%)	63 (67%)	10 (100%)
truncation	4 (100%)	19 (100%)	39 (92%)	10 (100%)
overflow	4 (100%)	35 (100%)	–	56 (89%)
stale	24 (100%)	–	–	–
enc.+trunc.	4 (100%)	18 (100%)	34 (99%)	7 (100%)

plus three standard deviations. Table 6 and Figure 3 show that trace checks alarm a median of 4 requests after each global fault and 24 requests after the partial stale fault, with 0.1% false alarms. MiniCheck needs 18–35 requests for global faults and rarely detects the stale one. The LLM verifier flags 75% of *clean* answers, so its alarm threshold is high; it needs 34–93 requests and misses two thirds of overflow deploys. The model's abstention rate is a reasonable signal for retrieval and truncation faults (7–10 requests), slow for overflow (56) and poor for stale documents.

4.4 RQ4: Regression tests from observed failures

We turned 35 observed cases into a regression suite: five clean questions that must stay correct and, for each fault, five failures that Diagnose repaired. Because every model response and support score is cached, the suite replays the full pipeline in about 7 seconds without loading a model; a code change that alters a prompt or a repair decision misses the cache and fails until it is re-recorded. To check that the suite has teeth, we disabled one diagnostic at a time and replayed it on the live model (Table 7). Every mutation failed exactly the cases of the fault it covers.

4.5 Sensitivity to scoring

The equivalence judge accepted aliases such as "Bob Iger" for "Robert A. Iger". We audited all 63 of its acceptances on the test set, overturned 13 (it had accepted a related entity, e.g. "Greene King" for "Bury St Edmunds"), and corrected them; 59 of 60 sampled rejections were right. With string matching alone, success under faults drops by 8–18 points but every conclusion holds: Diagnose beats no repair by +24.5 [19.5, 29.9] and Repair-all by +9.9 [5.7, 13.9], trace checks alone are within +1.3 [−0.9, 3.5], and self-check remains the worst policy.

Table 7: Regression suite under mutations of the repair logic (35 cases).

Mutation	Failing	Conditions
none	0	–
no truncation check	10	truncation, encoder+truncation
no low-retrieval check	8	encoder, encoder+truncation
no overflow check	5	overflow
no version check	4	stale
no re-check after repair	3	encoder, encoder+truncation

5 Lessons for Practitioners

Log what reaches the model. The most useful signals were comparisons the service could make with data it already had: rendered versus retrieved text, tokens sent versus tokens the server processed, and duplicate document identities in the context. Standard retrieval metrics measure what was retrieved, not what survived assembly; these checks cost nothing and alarmed within a few requests of a bad deploy.

Match the repair to the component. Re-asking the generator on the same broken context never helped, and doing every repair on every request made the serving fault worse. A repair needs to know which component failed.

Do not let a small model veto its own answers. The 3B model's verifier rejected most of its correct answers. Used as a gate, it destroyed good answers; used as a trigger for context repair, it cost little recovery but more damage than a text match. A frontier verifier is far better calibrated, but verification only decides *whether* to act; recovery comes from fixing the context.

Check again after each repair. When two faults co-occur, fixing the visible one may not be enough; re-checking support after a repair recovered several points under the encoder fault.

Treat abstention as a separate requirement. Repairs that add passages make the model less willing to say it does not know. None of the policies we tested achieved both high recovery and high abstention on unanswerable questions.

6 Threats to Validity

Construct. Faults are injected deterministically, and questions were filtered so that clean dense retrieval finds both supporting passages; real retrieval is noisier and real faults less tidy. The faults are modeled on reported incidents, but their severity (e.g., a 100-character truncation, a 512-token window) was chosen by us. The text-match support check works because HotpotQA answers are extractive spans; abstractive answers would need an entailment model.

Internal. The proposed policy's support check, MiniCheck's thresholds and the retrieval-confidence threshold were chosen on development or held-out calibration questions, disjoint from the test set, by rules fixed before the test run. On the test set

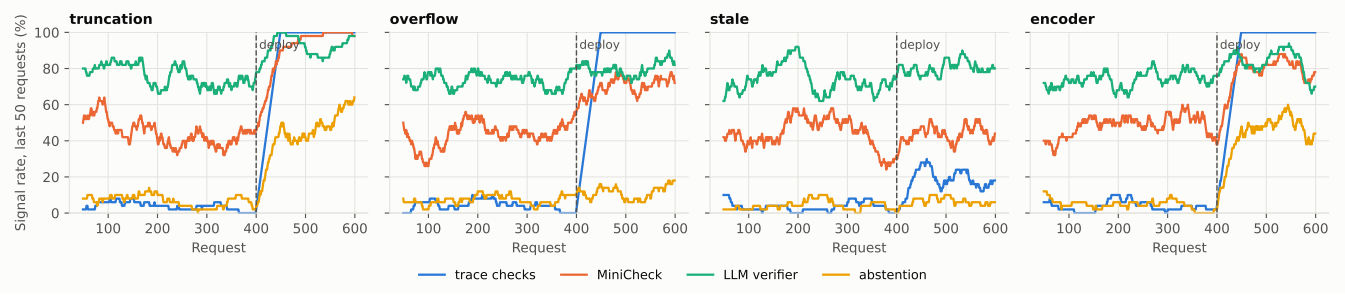


Figure 3: Signal rates in one simulated stream per fault; the dashed line marks the deploy.

the MiniCheck variant scored slightly higher than the proposed text-match variant (59.3% vs. 56.9%, difference not significant), which we report as is. Correctness relies partly on an LLM judge; all of its test acceptances were audited by the first author's AI assistant and corrected, and a blind audit sheet for two human annotators is part of the artifact.

External. We report one 3B model, greedy decoding, one dataset and one retriever. Larger models verify better: Claude Haiku 4.5 reached 0.77 balanced accuracy as a verifier on our answers, so self-check with a strong verifier would break fewer correct answers, although it still cannot recover missing evidence. We measured the strong verifier on a sample only, not as a full policy. Runs with Qwen2.5-7B and Gemma-2-9B are in progress with the same harness. Costs are measured on one laptop and are meaningful only relative to each other.

Conclusion validity. With 150 questions, pooled comparisons have intervals of roughly ±5 points; several per-condition differences between the stronger policies are not significant.

7 Conclusion

We injected six production-motivated faults into a RAG service and compared twelve runtime repair policies under a fixed call budget. Checks on what actually reached the model, computed from the service's own trace, recovered most of the lost accuracy, broke few correct answers, and detected faulty deploys within a handful of requests. Generic self-checking with the model's own verifier was worse than no repair, and blanket repair amplified a serving fault. The harness, data construction, recorded responses and regression suite are available for replication.

Artifact. Code, data construction scripts, recorded model responses, result files and the regression suite are in the project repository.

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